实验三：区块链数据分析

Lab 3: Blockchain Data Analysis

1. 实验目标 (Objectives)

- **理解数据组织 (Understand Data Organization):** 理解区块链作为公开账本的数据组织方式，掌握区块数据与费用数据的基本含义。
  Understand the data organization of blockchain as a public ledger, and grasp the basic meanings of block data and fee data.
- **掌握网络状态分析 (Master Network State Analysis):** 学会整理和分析链上区块信息，理解区块活跃度、资源使用情况与网络运行状态之间的关系。
  Learn to organize and analyze on-chain block information, and understand the relationship between block activity, resource utilization, and network operational status.
- **剖析费用机制 (Analyze the Fee Mechanism):** 围绕区块链费用机制，对连续区块中的费用数据进行统计与可视化分析。
  Focus on the blockchain fee mechanism to statistically analyze and visualize fee data across consecutive blocks.
- **培养数据表达能力 (Develop Data Articulation Skills):** 训练利用链上数据描述区块链网络状态的能力，并通过图表和文字分析形成基本的数据分析结论。
  Train the ability to describe the blockchain network state using on-chain data, and formulate basic data analysis conclusions through charts and text analysis.

2. 实验背景 (Background)

区块链不仅是交易执行和智能合约运行的平台，也是一个持续增长的公开账本。链上的每一个区块都记录了时间、交易数量、资源消耗和费用信息，这些数据能够客观反映网络在某一时间段内的运行状态。
Blockchain is not only a platform for transaction execution and smart contract operation, but also a continuously growing public ledger. Every block on the chain records time, transaction count, resource consumption, and fee information. These data can objectively reflect the operational state of the network during a specific period.

本实验要求你从数据分析的视角出发，对连续区块中的基础数据进行整理、汇总和分析。重点关注两类核心问题：一是区块活动情况，二是费用变化趋势。通过本实验，你应能够从原始链上数据中提取有意义的信息，并用直观的图表和严谨的结论对网络状态进行描述。 **注意：由于本实验需要分析 Base Fee 字**

段，实验应基于支持 EIP-1559 费用机制的以太坊网络（如以太坊主网或 Sepolia 等测试网）开展。
This lab requires you to approach from a data analysis perspective to organize, summarize, and analyze fundamental data from consecutive blocks. Focus on two core issues: first, block activity; second, the trend of fee changes. Through this lab, you should be able to extract meaningful information from raw on-chain data and describe the network state using intuitive charts and rigorous conclusions. **Note: Since this lab requires the analysis of the Base Fee field, the experiment should be conducted on an Ethereum network that supports the EIP-1559 fee mechanism (such as the Ethereum Mainnet or testnets like Sepolia).**

3. 实验内容 (Experiment Tasks)

任务一：区块数据汇总分析 (Task 1: Block Data Summary and Analysis)

收集最近一段连续区块的数据，并对其中至少 10 个区块进行详细的汇总分析。获取的每个区块至少应包含以下字段：
Collect data from a recent sequence of consecutive blocks, and conduct a detailed summary and analysis of at least 10 blocks. Each retrieved block must include at least the following fields:

- 区块高度 (Block Number)
- 出块时间 (Timestamp)
- 区块内交易数量 (Transaction Count)
- 费用接收地址（区块头 beneficiary，对应部分 RPC 返回字段 miner）(Fee Recipient: block header beneficiary, corresponding to the 'miner' field in some RPC responses)
- 消耗的 Gas (Gas Used)
- Gas 上限 (Gas Limit)

在此数据基础上，完成以下分析任务：
Based on this data, complete the following analysis tasks:

- 描述最近区块的活跃程度变化（如哪些区块交易数量较多，哪些较少）。
  Describe the changes in the activity level of recent blocks (e.g., which blocks have more transactions and which have fewer).
- 比较不同区块的资源使用情况，分析 Gas Used 占 Gas Limit 的比例（即区块填充程度）。
  Compare the resource utilization across different blocks, analyzing the ratio of Gas Used to Gas Limit (i.e., the block fill rate).
- 结合时间与区块数据，说明链上活动在短时间内是否存在明显的波动。
  Combine time and block data to explain whether on-chain activity shows significant short-term fluctuations.

- 对上述数据进行归纳，形成对当前网络基本状态的总体判断。
  Summarize the above data to form an overall assessment of the current basic network state.

任务二：费用机制分析 (Task 2: Fee Mechanism Analysis)

围绕一段连续区块中的费用数据，分析网络费用的宏观变化趋势。请收集**至少 20 个连续区块**的费用相关数据，重点关注 Base Fee、Gas Used 和 Gas Limit，并完成以下分析：
Focus on the fee data within a sequence of consecutive blocks to analyze the macro trend of network fees. Please collect fee-related data for **at least 20 consecutive blocks**, focusing on Base Fee, Gas Used, and Gas Limit, and complete the following analysis:

- 绘制 Base Fee 随区块高度变化的趋势图。
  Plot a trend chart of the Base Fee against the block height.
- 观察 Base Fee 的变化是否平稳，指出是否存在明显的上升或下降区间。
  Observe whether the Base Fee changes smoothly, and identify if there are distinct upward or downward intervals.
- 结合区块的填充程度（Gas Used 占比），分析 Base Fee 的调整与区块资源使用程度之间是否存在逻辑关联。
  In conjunction with the block fill rate (the ratio of Gas Used), analyze whether there is a logical correlation between the adjustment of the Base Fee and the block's resource utilization.
- 简要说明当前的费用机制反映出了网络怎样的拥堵程度或活跃程度变化。
  Briefly explain what the current fee mechanism reveals about changes in network congestion or activity levels.

4. 核心数据与分析要求 (Data and Analysis Requirements)

本实验不强调具体的数据获取实现方式（可自由选择 Web3 RPC、以太坊浏览器 API 或第三方数据平台），而强调是否能够围绕链上数据完成规范的整理与分析表达。在分析层面，应至少体现以下能力： This lab does not emphasize the specific implementation method for data acquisition (you may freely choose Web3 RPC, Ethereum Explorer APIs, or third-party data platforms). Instead, it emphasizes the ability to conduct standardized organization and analytical expression based on on-chain data. At the analytical level, you should demonstrate at least the following capabilities:

- 能够将原始、底层的区块字段整理为可读的结构化信息。
  The ability to organize raw, low-level block fields into human-readable, structured information.

- 能够通过专业的图表和简要的文字分析组织实验内容，做到图文并茂。
  The ability to organize experimental content using professional charts and brief text analysis, achieving a visually supported presentation.
- 能够基于客观数据对网络状态作出解释，拒绝单纯的字段罗列。
  The ability to explain the network state based on objective data, rather than merely listing raw fields.
- 能够从“区块活跃度”和“费用变化”两个独立又相关的角度，对链上运行情况进行立体说明。
  The ability to provide a multi-dimensional explanation of on-chain operations from two independent yet related perspectives: "block activity" and "fee changes."

5. 实验结果与提交要求 (Assessment)

请提交一份 PDF 格式的实验报告，内容必须包含以下两个部分：
Please submit a lab report in PDF format, which must include the following sections:

（1）图表展示 (Visualizations / Charts)

- 必须包含：**20 个连续区块**的 Base Fee 变化趋势图。
  Must include: A trend chart of the Base Fee changes across **20 consecutive blocks**.
- 必须包含：至少 1 个能够反映区块活跃度或资源使用情况的辅助图表（如：区块交易数量变化图、Gas Used 变化图、或 Gas Used 与 Gas Limit 的占比对比图）。
  Must include: At least 1 supplementary chart reflecting block activity or resource utilization (e.g., block transaction count trend, Gas Used trend, or a comparison chart of the Gas Used to Gas Limit ratio).

（2）分析结论 (Analytical Conclusions) 围绕实验中得到的图表与数据，给出结构化的分析结论，包括但不限于：
Based on the charts and data obtained in the lab, provide structured analytical conclusions, including but not limited to:

- 最近一段区块链网络活动整体是否稳定。
  Whether the recent blockchain network activity is generally stable.
- 区块交易数量是否存在明显波动。
  Whether there are significant fluctuations in the number of block transactions.
- Base Fee 的变化趋势特征（寻找显著的上升或下降区间）。
  Characteristics of the Base Fee trend (identifying notable upward or downward intervals).
- 区块资源使用情况与费用变化之间的逻辑关联。
  The logical correlation between block resource usage and fee changes.

- 你对该时间段内链上运行状态的总体判断总结。
  Your summary and overall judgment of the on-chain operational state during that period.